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Original article

Retro-mode imaging for the diagnosis of optic disc drusen: a case series



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ABSTRACT

Objective: We aimed to compare the detectability of optic disc drusen (ODD), using various non-invasive imaging techniques, including the novel retro-mode imaging (RMI), as well as to analyze the morphological characteristics of ODD on RMI.

Methods: This study involved seven patients with bilateral ODD, totaling 14 eyes. Multimodal imaging techniques, including multicolor fundus photography (MC), near-infrared reflectance (NIR), green and blue light fundus autofluorescence (G-FAF and B-FAF, respectively), and RMI were used to examine the eyes. FAF was used as the primary method of identifying ODD, and each method's detection rate was compared by two observers. Quantitative measurements of ODD included the number of ODD visualized by the RMI technique, the perimeter (P) and area (A) of ODD were identified.

Results: The average age of the patients included was 49.28 ± 23.16 years, with five of the seven being men. RMI was able to detect ODD in all cases, with a sensitivity of 100%, compared to MC (sensitivity 60.71%), NIR (sensitivity 60.71%), B-FAF (sensitivity 100%), G-FAF (sensitivity 100%). RMI was the only imaging technique capable of assessing ODD morphology and quantifying ODD.

Conclusions: RMI is a promising imaging modality for diagnosing superficial ODD, providing valuable information on the distribution, location, and size of ODD. We suggest the incorporation of RMI as a complementary tool for diagnosing and monitoring ODD in combination with other multimodal imaging methods.

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Imagen en retromodo para el diagnóstico de drusas del disco óptico: una serie de casos

R E S U M E N

Palabras clave:

Imagen de retromodo
Drusas del disco óptico
Imágenes multimodales
Pseudoedema de papila óptica

Objetivo: Nuestro principal objetivo es el de comparar la capacidad para detectar las drusas del disco óptico (ODD) utilizando diversas técnicas de imágenes no-invasivas, incluida la novedosa técnica de imagen de retro-modo (RMI). Como segundo objetivo analizamos las características morfológicas de las ODD bajo esta última técnica.

Materiales y métodos: Este estudio incluyó un total de siete pacientes con ODD bilaterales, obteniendo un total de 14 ojos analizados. Se utilizaron técnicas no-invasivas de imágenes multimodales, que incluyeron fotografía multicolor del fondo de ojo (MC), reflectancia en infra-rojo (NIR), autofluorescencia en luz verde y en luz azul (G-FAF y B-FAF, respectivamente), y RMI. La FAF se utilizó como el método principal para el diagnóstico de ODD. Dos observadores realizaron las comparaciones obteniendo las tasas de detección de cada uno de los métodos. Las mediciones cuantitativas de las ODD incluyeron el número, el perímetro (P) y el área (A) de las ODD identificadas mediante la técnica de RMI.

Resultado: La edad promedio de los pacientes incluidos fue de 49.28 ± 23.16 años, cinco de los siete pacientes fueron de sexo masculino. La técnica de RMI pudo detectar ODD en todos los casos, con una sensibilidad del 100%, en comparación con MC (sensibilidad del 60.71%), NIR (sensibilidad del 60.71%), B-FAF (sensibilidad del 100%), G-FAF (sensibilidad del 100%). RMI fue la única técnica de imagen capaz de evaluar morfológica y cuantitativamente las ODD.

Conclusiones: RMI es una prometedora modalidad no-invasiva de imagen para diagnosticar ODD superficiales, proporcionando información valiosa sobre la distribución, ubicación y tamaño de estas, por lo tanto, mediante nuestros resultados sugerimos la incorporación de la novedosa técnica de RMI como una herramienta complementaria para el diagnóstico y seguimiento de ODD en combinación con los otros métodos de imagen multimodales.

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Introduction

Optic disc drusen (ODD) are described as acellular deposits located within the optic nerve, intracellularly and extracellularly, and may calcify over time.¹ These deposits are situated above the lamina cribrosa and below the Bruch membrane plane.²

ODD is typically asymptomatic, although some cases have reported serious visual field defects and vascular obstructive complications.³⁻⁶

The condition is frequently discovered by chance during a routine eye exam and is not typically associated with any ocular or systemic disease. ODD is bilateral in 75% of cases and has an estimated prevalence ranging from 0.3% to 2% of the general population, with no sex predilection.⁷

Although superficial ODD often mimics papilledema and should be distinguished from true papilledema arising from more dangerous causes using different auxiliary tests, including fundus autofluorescence (FAF), optical coherence tomography (OCT), fluorescein angiography (FA), B-scan ultrasound.⁸

Introducing new high-resolution multimodal imaging modalities has enabled ODD to be detected at much earlier stages and deeper localizations than possible, thus improving research into their underlying mechanisms.⁹

Retro-mode imaging (RMI) is a novel modality that captures the reflectance of infrared laser light emitted by a confocal scanning laser ophthalmoscope (cSLO) using a sideways deflected confocal aperture with a central stop. This method allows the creation of a shadow to acquire high-resolution pseudo-3-dimensional images from the deep retinal structures. It is beneficial in studying deep retinal pathologies and retinal pigment epithelium (RPE) changes.¹⁰

This study aimed to compare the detectability of ODD using various non-invasive imaging techniques, including RMI, as well as to describe and quantify ODD in RMI.

Material and methods

This was a retrospective study of seven patients diagnosed with bilateral ODD evaluated at the Department of Ophthalmology of the Intercommunal Hospital of Creteil (Creteil, France) between September 2022 through Mars 2023. This study conformed to the tenets of the Declaration of Helsinki.

Data collected included demographic information and all patients underwent a complete ophthalmological examination, including the measurements of the best-corrected visual acuity (BCVA) using a Snellen chart, binocular indirect ophthalmoscopy, and slit lamp examination.

Table 1 – Patient demographic characteristics and their pertinent ocular history.

Patient	Age (Years)	Sex	Best Corrected Visual Acuity (Snellen)	Ocular History
1	18	M	20/20 OD 20/20 OS	None
2	58	M	20/40 OD 20/80 OS	Bilateral cataract surgery, bilateral panretinal photocoagulation for proliferative diabetic retinopathy (2015)
3	71	F	20/20 OD 20/25 OS	Bilateral cataract surgery
4	82	M	20/200 OD 20/400 OS	Bilateral cataract surgery, dry age-related macular degeneration (diagnosed in 2003)
5	52	M	20/20 OD 20/20 OS	None
6	35	F	20/20 OD 20/20 OS	None
7	29	M	20/20 OD 20/20 OS	None

Retinal examinations using the multimodal imaging system Mirante cSLO/OCT (Nidek, Japan), included fundus multicolor (MC) photography, near-infrared reflectance (NIR), FAF using green (G-FAF) and blue light (B-FAF), and RMI. The latter was performed with different laterality settings (right and left) with the respective confocal scanning laser ophthalmoscope (cSLO) photography.

On fundus examination, ODD is suspected when ill-defined or blurred papillary borders are present, together with a pale coloration, and morphologic changes of the optic nerve head. Diagnosis of ODD relied primarily on FAF imaging. On FAF, ODD appears as round or oval hyper autofluorescent structures with irregular edges. ODD at distinct depths exhibit varying degrees of hyperautofluorescence.

Spectral-domain optical coherence tomography (SD-OCT) was performed on all patients using Mirante (Nidek, Japan) and all patients underwent also B-scan ocular ultrasound (Quantel Medical, France). Both SD-OCT and ultrasound were performed to support the diagnosis of ODD.

In SD-OCT, **superficial ODD appears with an external hyperreflective border**, and a central hyporefective area, and exhibits a posterior shadowing effect. Moreover, SD-OCT with enhanced-depth imaging was used to differentiate and subsequently exclude patients with peripapillary hyper-reflective ovoid mass-like structures (PHOMS). On ultrasound, ODD corresponds to increased optic nerve head reflectivity.

All ophthalmic images were stored and reviewed using a software imaging database (Mirante, Nidek, Japan). Two observers analyzed all images (G.R and E.C). The interobserver agreement was tested by Cohen Kappa analysis.

For the quantitative analysis of ODD in terms of area (A) and perimeter (P), we used Mirante (Nidek, Japan) functional software that allowed us to obtain A and P individually, for each of the ODD, using an image enlargement and a caliper for the measurements.

Statistical analysis

Statistical analysis was performed using SPSS version 26 (IBM, IL, USA) including descriptive statistics for main clinical features. The interobserver agreement was tested by Cohen Kappa analysis. The significant value was under $P < 0.05$.

Table 2 – Summarizes the detection rates of ODDs using each of the non-invasive imaging techniques.

Imaging Modality	Observer 1 ODD Detected = n%	Observer 2 ODD Detected = n%
Multicolor (MC)	9/14 (64.28 %)	8/14 (57.14 %)
Near-Infrared (NIR)	9/14 (64.28 %)	8/14 (57.14 %)
G-FAF	14/14 (100 %)	14/14 (100 %)
B-FAF	14/14 (100 %)	14/14 (100 %)
Retro-mode imaging (RMI)	14/14 (100 %)	14/14 (100 %)

Results

We identified ODD in 14 eyes of seven patients. For each included eye, the optic disc was hyperautofluorescent, consistent with a diagnosis of ODD. Five of the 7 patients were men, and the mean age was 49.28 ± 23.16 years, with a range of 18–82 years.

Central visual acuity was well preserved in five patients, with visual acuity ranging from 20/20 to 20/25 (Snellen). One patient had a history of proliferative diabetic retinopathy treated by laser panphotocoagulation, whose visual acuity was 20/40 in the right eye and 20/80 in the left eye. One patient had a concomitant diagnosis of dry age-related macular degeneration (AMD) with visual acuity of 20/200 and 20/400 in the right and left eye, respectively. Mean BCVA on the 14 included eyes was $20/27 \pm 20/54$ (0.13 ± 0.43 LogMAR average). The characteristics of the patients are shown in [Table 1](#).

Detection of ODD using non-invasive multimodal imaging

MC identified 60.71% (ranging from 57.14% to 64.28%) of cases, where ODD appeared as highly refractive whitish-yellow bodies, and the optic discs were often crowded and elevated. NIR reported the same percentage of ODD detection. G-FAF, B-FAF, and RMI demonstrated the highest diagnostic sensitivity (100%) for ODD detection, as shown in [Table 2](#).

The RMI technique detected ODD using right and left deviation scans in all 14 eyes and demonstrated morphological characteristics illustrated by pseudo-3-dimensional images with oval reliefs on the contour of the optic nerve head, as well as white refractive dots that could be rapidly individualized,

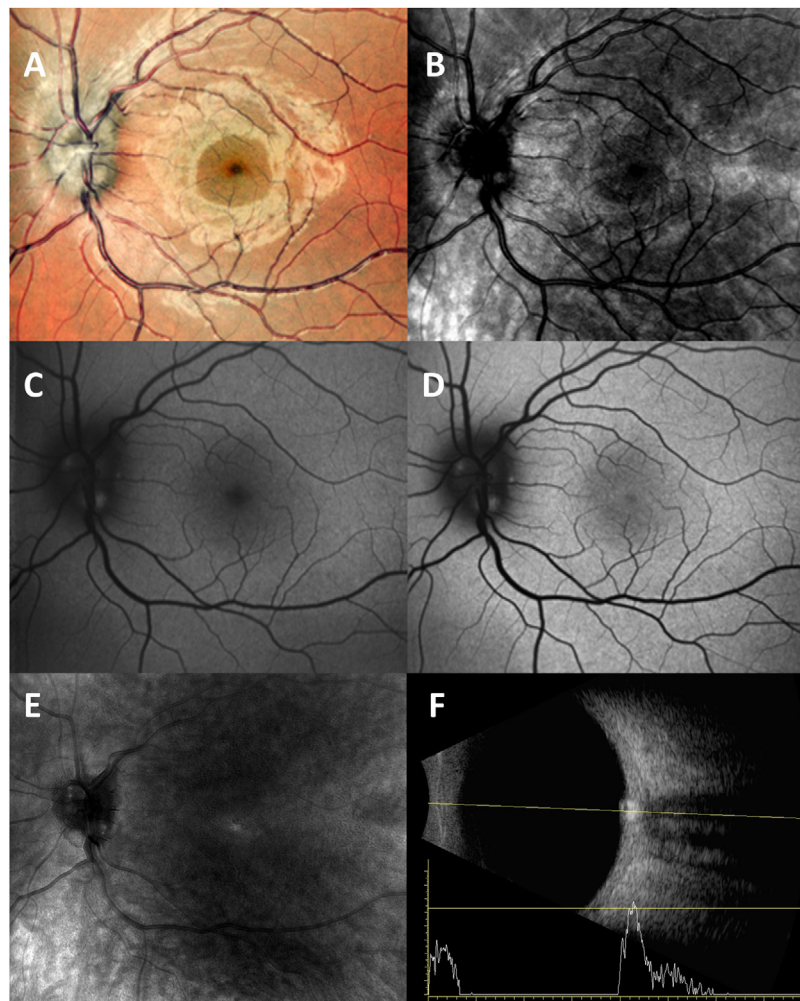


Fig. 1 – Multimodal imaging by Mirante (Nidek, Japan) realized on 18-year-old men. Multicolor fundus photographs (MC) of the left eye (a) show an optic nerve with elevation and erasure of the edges; Near-infrared reflectance (NIR) (b) does not demonstrate any alteration; Blue light autofluorescence (B- FAF) (c) and green light autofluorescence (G-FAF) (d) showed the location of 2 well-defined hyperautofluorescent spots (nasal and lower). Retro-mode imaging (RMI) using the deviation left (e) shows a high relief in the areas mentioned above in the autofluorescence in conjunction with refractive points. B-scan ultrasound (f) showing increased reflectivity from the optic nerve head consistent with calcified ODD.

coinciding with surface calcifications subsequently confirmed with SD-OCT imaging.

With regards to the location of ODD, they were most frequently observed in the nasal-superior quadrant, followed by the temporal-superior quadrant, nasal-inferior quadrant, and temporal-inferior quadrant, with detection rates of 78.57% (11/14), 50% (7/14), 42.85% (6/14), and 35.71% (5/14), respectively (Cohen Kappa analysis of interobserver agreement, $\kappa = 0.86$, $P = 0.025$).

Cases description

Case #1

An 18-year-old male was referred to our clinic with suspected asymptomatic bilateral papilledema, with no significant medical or ocular history. The anterior segment examination revealed no abnormalities in either eye and the best corrected visual acuity was 20/20 in both eyes. Fundus examination

revealed a mild elevation with blurred optic disc margins but no vessel obscuration. To further investigate the case, multimodal imaging was performed and presented in Fig. 1.

Fundus MC photography left eye (Fig. 1a) showed protrusion and the optic disc margin blurring without other visible retinal alterations. NIR (Fig. 1b) was unremarkable. FAF in B-FAF (Fig. 1c) and G-FAF (Fig. 1d) presented a heterogeneous signal containing hyperautofluorescent points very well defined in the nasally and inferior sector of the optic nerve.

By introducing the RMI technique (Fig. 1e), we performed both right and left deviations, and the latter showed the highest quality and sensitivity in the image. We observed the optic disc drusen (ODD) speckled gray color on the optic disc and lighter-gray peripapillary relief on the hyperreflective points located in the nasal and inferior region of the optic nerve, which FAF previously detected.

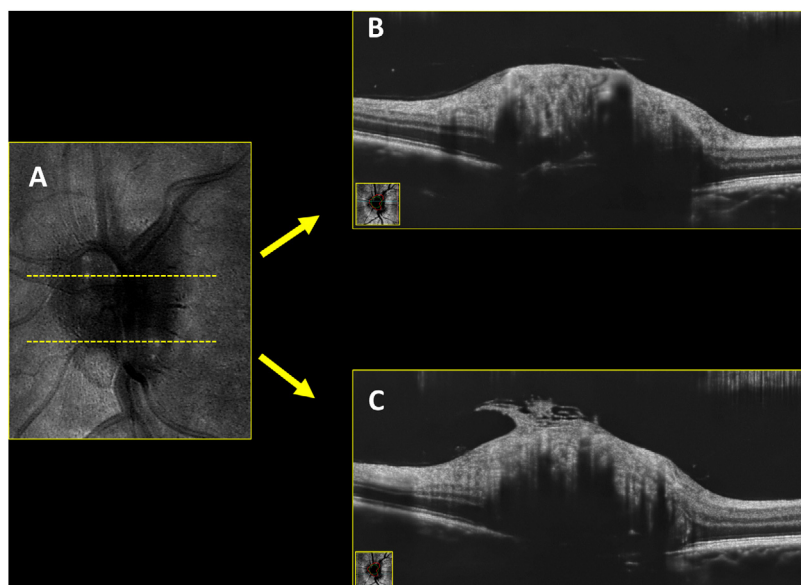


Fig. 2 – Retro-mode imaging (RMI) using left deviation (a) in the left eye showing hyperreflective white dots, under spectral-domain optical coherence tomography (SD-OCT) scan in the nasally dot (b) and in the inferior dot (c) we find a hyperreflective sign that corresponds to a calcified drusen.

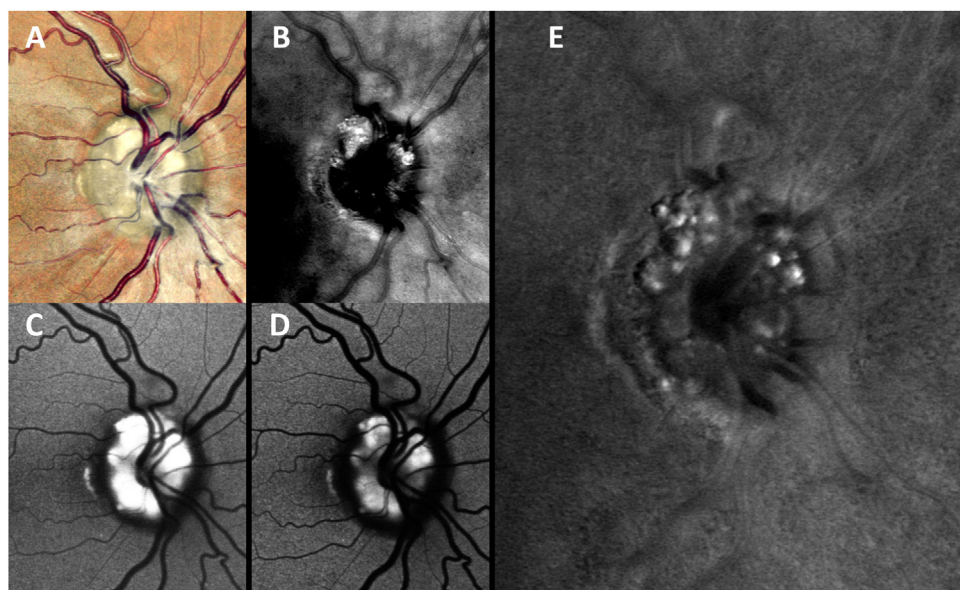


Fig. 3 – Multimodal imaging performed on 29-year-old men. The MC image (a) showed an elevation of the papilla along with superior whitish coloring, NIR (b) presented superior hyperreflectance, B-FAF (c) showed hyperautofluorescence with no detail throughout the optic nerve, G-FAF (d) better detailed hyperautofluorescence and RMI (e) illustrated a high morphological visualization detection of ODD.

B-scan ultrasound (Fig. 1f) of the left eye showed increased reflectivity from the optic nerve head consistent with calcified ODD.

Further, ODD was evaluated using RMI and SD-OCT scan images of the optic nerve head (Fig. 2), which showed calcification of the ODD as hyperreflective lines on an ovoid mass with elevated irregular disc borders and thinning of the retinal nerve fiber layer. The patient, who was asymptomatic, was advised to have further follow-up examinations.

Case #2

A 29-year-old hyperopic man presented for a routine ophthalmic examination and glasses prescription. Visual acuity under correction was 20/20 on the right eye and 20/20 on the left eye.

Multimodal imaging was performed on the right eye (Fig. 3), MC (Fig. 3a) illustrated an optical nerve head with undefined borders and with an over-elevated appearance with a pale coloring in the upper part of the papilla, NIR (Fig. 3b) showed a diffuse gray color at the top of the disk with several light gray

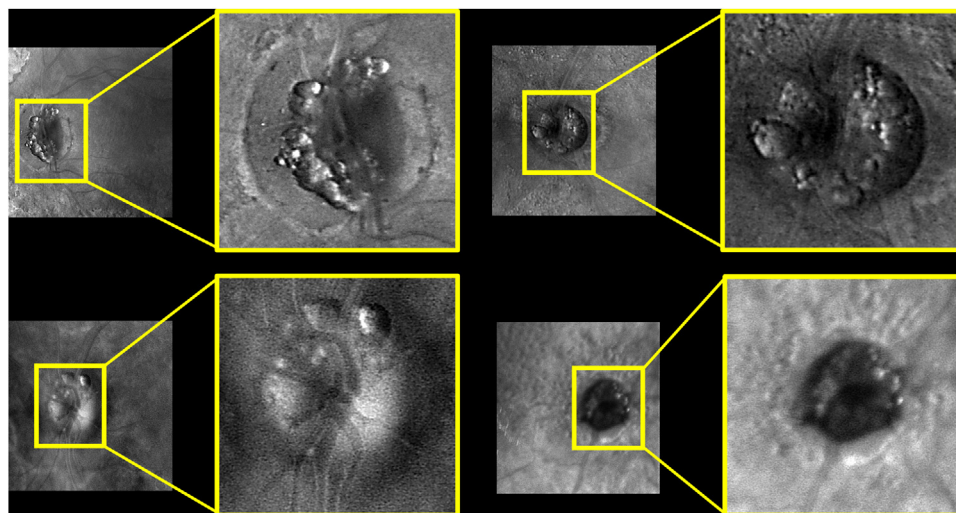


Fig. 4 – shows the display of ODD from 4 different patients using RMI, which demonstrated the possibility of observing by magnification the high resolution in the distribution and morphology of ODD.

to white spots, B-FAF (Fig. 3 c) illustrated a homogeneously hyperautofluorescent bright signal throughout the optic disc, G-FAF (Fig. 3d) demonstrated a have heterogeneous signal, ranging from the dim brightness of large, central conglomerate of ODD to the intense, punctate brightness.

RMI technique using left deviation (Fig. 3e) showed a grayish color with white stippling and a distinct, raised, ovoid morphology of the ODD with superior location over the optic nerve head. Our patient was recommended for continued follow-up.

Discussion

Optic disc drusen (ODD) is a clinical condition that is often misdiagnosed as papilledema due to similarities in presentation, such as bilateral elevated and blurred disc margins. Superficial ODD can be identified through the presence of yellow deposits on ophthalmoscopic examination. However, diagnosing ODD can be challenging with only fundoscopic examination or imaging, as ODD may not always be visible. On the other hand, buried ODD may be mistaken for optic disc swelling and require additional radiological imaging and invasive diagnostic techniques to distinguish them from more severe conditions such as intracranial tumors and increased intracranial pressure.^{1,2} ODD can be detected through various multimodal imaging methods, including multicolor (MC), red-free fundus photography, autofluorescence (FAF), and optical coherence tomography (OCT) imaging.¹¹

Our study found that RMI had the same high detection rate as FAF for the diagnosis of ODD. Furthermore, it showed a high interobserver agreement ($k=0.86$). G-FAF and B-FAF had the same detectability as RMI in all our cases, although individualizing each lesion was difficult as drusen cannot be easily identified in the vast majority of cases. G-FAF images showed better clarity of ODD than B-FAF images, as the latter shows a homogeneous bright signal throughout the optic disc, which was also reported by Yan et al.,¹² where they found

FAF highly sensitive for ODD detection. However, FAF mainly detects superficial drusen and does not provide morphological information.

The current recommendation for ODD visualization is enhanced depth imaging optical coherence tomography (EDI-OCT). The latter exhibits variably sized masses surrounded by a hyperreflective border. The improved penetrance afforded by EDI-OCT technology now enables quantification of drusen size, delineation of drusen borders, and assessment of the integrity of adjacent retinal structures.^{13,14}

The RMI technique, by creating an illusion of depth, allows highlighting the relief associated with pathological structures in the retina. This topographic information in pseudo-3-dimensional format enables delineation, quantification, morphological visualization, and precise identification of the location of each superficial ODD when compared to other flat-plane detection techniques like FAF.

The observed performance in the RMI for superficial ODD appears to demonstrate an interesting and high-quality technique for quantifying ODD, thereby revealing their topography and distribution, as seen in Fig. 4.

We observed that the RMI technique provided rapid visualization of superficial ODD through pseudo-3-dimensional image rendering, identifying the boundaries of each individual ODD based on a topographic relief map. Additionally, it detected superficial calcifications, as shown in Fig. 5.

Additionally, RMI, with its software that allows measuring the P and A of each ODD individually, appears to be a valuable technique for long-term monitoring of morphological changes in each of the ODD. Therefore, RMI can be a useful tool in the diagnosis and management of ODD, especially in cases where fundus photography and FAF imaging have limited diagnostic value.

Since 2009, the RMI technique has been used to investigate fundus diseases and has shown promise in the diagnosis and treatment of various retinal conditions, such as cystoid macular edema (CME), retinoschisis, polypoidal choroidal vasculopathy (PCV), central serous chorioretinopathy (CSCR),

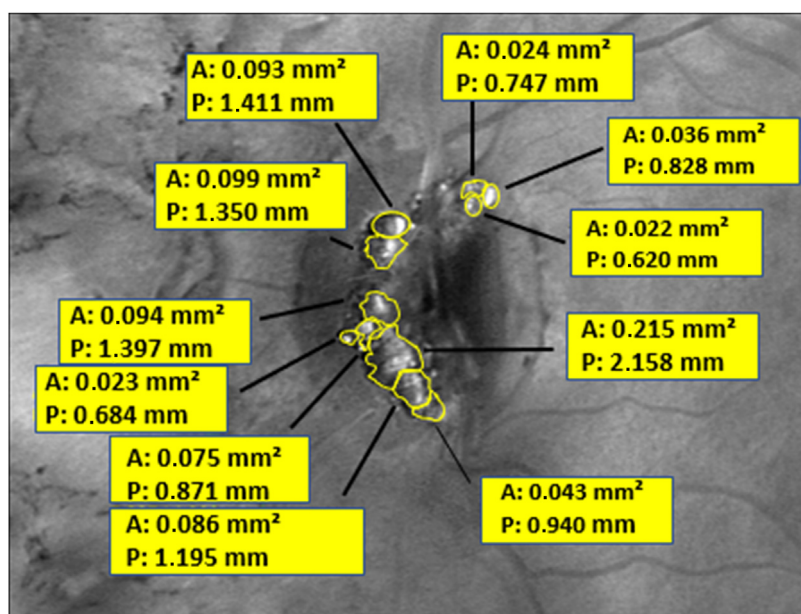


Fig. 5 – Measurement of area (A) and perimeter (P) of individuality optic disc drusen using Mirante (Nidek, Japan) on the RMI.

exudative age-related macular degeneration and diabetic retinopathy (DR), among others described in the literature.^{15–19} This plays an important role not only in enabling the analysis and detection of ODD.

In conclusion, our findings suggest that RMI is a promising imaging modality for detecting and managing ODD, which could ultimately improve outcomes for affected patients. Awareness of various patterns observed in multimodal imaging can enhance comprehension of the fundamental mechanisms and provide a more perceptive method of detecting ODD.

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Competing interest statement

The authors have no competing interest in publishing the present work.

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REFERENCES

- Chang MY, Pineles SL. Optic disc drusen in children. *Surv Ophthalmol*. 2016;61(6):745–58, <http://dx.doi.org/10.1016/j.survophthal.2016.03.007>.
- Spencer WH. XXXIV Edward Jackson Memorial Lecture: drusen of the optic disc and aberrant axoplasmic transport. *Ophthalmology*. 1978;85(1):21–38, [http://dx.doi.org/10.1016/s0161-6420\(78\)35696-7](http://dx.doi.org/10.1016/s0161-6420(78)35696-7).
- Lorentzen SE. Drusen of the optic disk. A clinical and genetic study. *Acta Ophthalmol (Copenh)*. 1966; Suppl 90:1–180.
- Lee KM, Hwang JM, Woo SJ. Hemorrhagic complications of optic nerve head drusen on spectral domain optical coherence tomography. *Retina*. 2014;34(6):1142–8, <http://dx.doi.org/10.1097/IAE.0000000000000040>.
- Romero J, Sowka J, Shechtman D. Hemorrhagic complications of optic disc drusen and available treatment options. *Optometry*. 2008;79(9):496–500, <http://dx.doi.org/10.1016/j.optm.2008.01.020>.
- Hamann S, Malmqvist L, Costello F. Optic disc drusen: understanding an old problem from a new perspective. *Acta Ophthalmol*. 2018;96(7):673–84, <http://dx.doi.org/10.1111/aos.13748>.
- Sahin A, Cingü AK, Ari S, Cinar Y, Çaça I. Bilateral optic disc drusen mimicking papilledema. *J Clin Neurol*. 2012;8(2):151–4, <http://dx.doi.org/10.3988/jcn.2012.8.2.151>.
- Palmer E, Gale J, Crowston JG, Wells AP. Optic nerve head drusen: an update. *Neuroophthalmology*. 2018;42(6):367–84, <http://dx.doi.org/10.1080/01658107.2018.1444060>.
- Jia X, Bao T, Wang S, Jiang T, Zhong Z, Zang Y, et al. Diagnostic Value of Systematic Imaging Examination in Embedded Optic Disc Drusen in Adolescents with Mild Visual Impairment. *J Ophthalmol*. 2020;2020:6973587, <http://dx.doi.org/10.1155/2020/6973587>.
- Mainster MA, Desmettre T, Querques G, Turner PL, Ledesma-Gil G. Scanning laser ophthalmoscopy retroillumination: applications and illusions. *Int J Retin Vitre*. 2022;8:71, <http://dx.doi.org/10.1186/s40942-022-00421-0>.
- Gise R, Gaier ED, Heidary G. Diagnosis and imaging of optic nerve head drusen. *Semin Ophthalmol*. 2019;34(4):256–63, <http://dx.doi.org/10.1080/08820538.2019.1620804>.
- Yan Y, Ludwig CA, Liao YJ. Multimodal imaging features of optic disc drusen. *Am J Ophthalmol*. 2021;225:18–26, <http://dx.doi.org/10.1016/j.ajo.2020.12.023>.
- Malmqvist L, Bursztyn L, Costello F, Digre K, Fraser LA, Fraser C, et al. The optic disc drusen studies consortium recommendations for diagnosis of optic disc drusen using optical coherence tomography. *J Neuroophthalmol*.

- 2018;38(3):299-307,
<http://dx.doi.org/10.1097/WNO.0000000000000585>.
14. Silverman AL, Tatham AJ, Medeiros FA, Weinreb RN. Assessment of optic nerve head drusen using enhanced depth imaging and swept source optical coherence tomography. *J Neuroophthalmol*. 2014;34(2):198-205, <http://dx.doi.org/10.1097/WNO.0000000000000115>.
15. Corradetti G, Byon I, Corvi F, Cozzi M, Staurengi G, Sadda SR. Retromode illumination for detecting and quantifying the area of geographic atrophy in non-neovascular age-related macular degeneration. *Eye*. 2022;36:1560-6.
16. Giansanti F, Mercuri S, Serino F, Caporossi T, Savastano A, Rizzo C, et al. Scanning laser ophthalmoscopy retromode imaging compared to fundus autofluorescence in detecting outer retinal features in central serous chorioretinopathy. *Diagnostics*. 2022;12:2638.
17. Shin YU, Lee BR. Retro-mode Imaging for retinal pigment epithelium alterations in central serous chorioretinopathy. *Am J Ophthalmol*. 2012;154:155-63, e4.
18. Su Y, Zhang X, Wu K, Ji Y, Zuo C, Li M, et al. The non-invasive retro-mode imaging of confocal scanning laser ophthalmoscopy in myopic maculopathy: a prospective observational study. *Eye*. 2014;28:998-1003.
19. Lopez JM, Rabinovic M, Mehanna CJ, Ricciotti G, Colantuono D, Souied EH. Acute zonal occult outer retinopathy (AZOOR): case report with confocal scanning laser ophthalmoscopy retromode imaging. *J Fr Ophtalmol*. 2023;S0181-5512(23):386-8, <http://dx.doi.org/10.1016/j.jfo.2023.08.001>.